

Conceptual End-to-End Data Pipeline for Beejan Technologies

Beejan Technologies currently receives customer complaints through social media, call centre logs, SMS, and website forms. But they are faced with challenges, including:

1. The sources produce data in different formats and at different frequencies.
2. The organisation has no single pipeline.
3. The data is stored separately, so the reporting team often have to manually compile many spreadsheets before they can produce their reports, and this delays reports.
4. Teams also work in silos.

This conceptual end-to-end pipeline addresses this problem by creating a centralised workflow through which complaints can be collected from the different sources, ingested, transformed and made available for analysis and different teams in need of it. My design thinking is guided by these seven stages of data engineering, which are:

1. Source Identification: The data sources are social media platforms, call centre systems, SMS or customer messaging systems, and website forms. These sources are expected to differ significantly in structure and frequency.

2. Ingestion Strategy: A hybrid ingestion strategy is appropriate because of the variety of sources. Social media and SMS may generate individual complaints continuously; hence, they can be **streamed** via **API Ingestion**, while call-centre systems may produce files containing large numbers of records periodically; hence, they can be ingested in **batches** via **file upload**. Website forms may produce individual submissions that can be captured immediately or in small batches; hence, complaints exported from the website's database can be ingested via **file uploads**, or they can be directly streamed

3. Processing/Transformation: Raw customer complaints will need to be processed into a state ready for analysis. The processing layer therefore performs several activities. The data will need to be **validated, cleaned, standardised, normalised, and de-duplicated**. Sensitive customer information will need to be protected. Personally identifiable information (PII) will need to be masked and appropriate access restrictions put in place.

4. Storage: I propose a combination of a **data lake and data warehouse**. The raw data will be in the data lake preserving the original records received from the different sources. This is important for traceability, auditing, and future reprocessing. The structured, business-ready datasets for reporting and analytical workloads will be in the data warehouse.

5. Serving: This is where the processed data is made available for use by the business. A dashboard can provide management with a near-real-time or regularly refreshed view of

complaint activity. Reporting teams should be able to query curated complaint data through a consistent analytical interface. If the business wants, ticketing platforms can be integrated.

6. Orchestration and Monitoring: The orchestration will define dependencies between these stages and files and determine when each process should execute. Real-time flows should process events continuously, while batch processes can run hourly or daily depending on business requirements. The monitoring layer will track pipeline failures, processing latency, data freshness, record counts, and data-quality checks. If any part of the steps fail unexpectedly, the responsible team should receive an alert.

7. DataOps: The pipeline will have separate development, testing, and production environments. Changes will be version-controlled and tested before being deployed to the production environment. Automated quality checks should prevent obviously incorrect or incomplete datasets from reaching the reporting layer. Access to customer information will follow appropriate security principles.

Assumptions: Some of the assumptions made are that **Beejan** has appropriate access to the data sources and the relevant systems provide APIs, exports, webhooks, or another mechanism through which data can be extracted. It is also assumed that each complaint contains enough information to identify the source and timestamp and a customer identifier. Lastly, I assume the organisation has access to a scalable cloud environment and requires both analytical reporting and operational visibility.

Potential Challenges: One major challenge will be **data inconsistency**. Different channels may represent customers, dates, complaint types, and locations differently. Another challenge is **data volume and velocity**. A major network outage or service disruption could result in a sudden spike in complaints. **Data privacy** is another significant concern because complaints may contain names, phone numbers, account information, or other sensitive details. **Complaint categorization** may also be imperfect. Customers may describe the same problem using very different language.

Unknowns: Some of the unknowns are the exact volume and velocity of complaints from each source, whether a reliable customer identifier exists across all channels. Does the business have official complaint categories? Finally, integration with operational systems such as ticketing or customer-service platforms should be clarified.